



GOVERNMENT OF THE PEOPLE'S REPUBLIC OF BANGLADESH
MINISTRY OF DISASTER MANAGEMENT AND RELIEF (MoDMR)

Department of Disaster Management (DDM) • Cyclone Preparedness Programme (CPP)

TURNKEY NATIONAL TECHNICAL SPECIFICATION & INVESTMENT PROPOSAL

GREENSIGNAL AI

National Automated Disaster Response, Hydrometric Early Warning
& Strategic Resource Allocation Infrastructure

A Unified Multi-Agency Sovereign Ecosystem Integrating 2,500+ Automated Hydrometric Telemetry Stations, 495 Upazila Disaster Operations Hubs, 1,250 Climate-Hardened Union Relief Silos, and Resilient Multi-Million User Digital Command Applications.

OFFICIAL PROJECT CONTROL & REGISTRATION RECORD

Project Title:	GreenSignal AI: National Automated Disaster Response Platform
Project Code / Ref:	MoDMR-NDMP-2026-GS01 / GOB-Turnkey-P1
Total Estimated Outlay:	BDT 100,00,00,000 (BDT 100.00 Crore / Approx. USD 84.5M)
Target Project Lifecycle:	36 Months Turnkey Deployment + 36 Months Comprehensive O&M
Sponsoring Ministry:	Ministry of Disaster Management and Relief (MoDMR)
Lead Implementing Agency:	Department of Disaster Management (DDM) & GreenSignal Technical Consortium
Strategic Key Partners:	CPP, BMD, FFWC (BWDB), SPARRSO, BTRC, Armed Forces Division (AFD)
Security & Data Classification:	Official Public Safety Critical / Sovereign National Infrastructure
Submission Date:	September 2026 • Edition 3.0 (Comprehensive National Release)

SUBMITTED BY:

National Technical Architecture Consortium

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Authorized Engineering Seal & Signature

SPONSORED & ENDORSED BY:

Department of Disaster Management (DDM)

Joint Secretary / Director General
Ministry of Disaster Management and Relief (MoDMR)
Bangladesh Secretariat, Dhaka

Official Ministry Review & Verification Stamp

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1.1 Strategic Humanitarian & Geographic Imperative

Bangladesh is universally recognized as one of the world's most climate-vulnerable deltaic nations. Spanning over 230 transboundary rivers, the Meghna, Jamuna, and Padma basins funnel an astronomical volume of monsoonal discharge into the Bay of Bengal, resulting in annual inundations of 20% to 68% of the national territory. Furthermore, the 710 km coastline is repeatedly impacted by intense cyclonic storm surges, while the northeastern Haor wetlands face devastatingly rapid flash floods that submerge standing crops in fewer than 6 hours.

1.2 The "Last-Mile Gap" in Conventional Disaster Management

While macroscopic forecasting by the Bangladesh Meteorological Department (BMD) and hydrologic modeling by the Flood Forecasting and Warning Centre (FFWC) have drastically reduced cyclonic fatalities over past decades, an acute operational gap persists between macro-level meteorological alerts and ground-level humanitarian execution. Historical response challenges include:

- **Manual Telemetry Latency:** River gauges and rainfall data in remote upazilas often rely on physical chalk boards and manual phone calls, resulting in 2- to 4-hour lags before data reaches the national command center.
- **Static Relief Allocation:** Relief food, water purification tablets, and medical kits are frequently dispatched based on outdated census figures rather than live disaster displacement density.
- **Telecommunications Vulnerability:** Severe Category-4 cyclone landfalls invariably sever commercial electrical grids and knock down cellular base transceiver stations (BTS), isolating offshore islands.
- **Relief Distribution Leakage:** Paper-based relief manifests lack real-time digital verification, leading to misdirection, duplicated relief distribution, and stock-outs in isolated rural unions.

1.3 The GreenSignal AI Paradigm Shift

GreenSignal AI bridges this gap through a unified sovereign ecosystem combining automated industrial IoT telemetry, edge artificial intelligence, decentralized satellite/LoRaWAN fallback networks, and military-grade logistical dispatch. By synchronizing 2,500+ automated hydrometric stations directly with 495 Upazila Command Hubs and 1,250 Union Relief Buffer Silos, the platform automates the sensor-to-shelter decision loop within 30 seconds of an impending flood peak or cyclone surge.

NATIONAL KEY PERFORMANCE TARGETS (36-MONTH DELIVERY)	
Early Warning Latency:	< 30 Seconds from sensor threshold breach to automated cell broadcast & volunteer SMS.
Ground Truth Telemetry:	2,500+ solar-powered stations with 21-day autonomy during complete cloud submersion.
Grassroots Volunteer Mobilization:	Seamless offline PWA connectivity for 76,000+ CPP leads & 2.5M registered responders.
Strategic Relief Integrity:	100% auditable QR/NID biometric verification across all 1,250 Union Relief Buffer Silos.
System Availability SLA:	99.999% uptime via Tier-IV National Data Center with hot active-active DR failover.

SECTION 2.0: MULTI-TIER INSTITUTIONAL & OPERATIONAL ARCHITECTURE

Disaster response in Bangladesh requires harmonious coordination across multiple administrative tiers—from the central Cabinet crisis room down to remote coastal island volunteers. GreenSignal AI establishes an integrated digital and physical hierarchy designed to align with the National Plan for Disaster Management (NPDm).

2.1 Multi-Tier Stakeholder Integration Matrix

Operational Tier	Primary User Groups	Platform Interface	Core Operational Responsibilities
Tier 1: National Ministerial (Cabinet & DDM)	Minister, Secretary, DDM Director General, Armed Forces Division (AFD)	National Macro Command Portal (Dual 8K Central War-Room Display)	- Inter-district strategic grain/medical transfers - National state of emergency declarations - Armed Forces helicopter airlift mobilization - International humanitarian aid tracking
Tier 2: Tactical District & Upazila (DDMC / UZDMC)	Deputy Commissioners (DC), UNOs, Upazila Project Implementation Officers (PIO)	Tactical GIS Command Suite (public/index.html & district-planner)	- Real-time upazila hazard mapping & sensor telemetry - Dynamic shelter quota & food allocation balancing - Transport fleet routing & rescue boat dispatch - Direct responder messenger dispatching
Tier 3: Grassroots Field Responders (Union & Village)	76,000+ CPP Unit Leads, Red Crescent (BDRCS), Ansar-VDP, Community Health Workers	Field Worker PWA Mobile App (Offline-first IndexedDB / 2G sync)	- Door-to-door siren evacuation & roll-calls - GPS-tagged damage & casualty distress reporting - Relief distribution verification (NID/QR scan) - Survivor medical triage logging & SOS beacons
Tier 4: Affected Citizens (General Population)	Rural villagers, coastal islanders, urban slum dwellers, farmers	Public Cell Broadcast SMS, IVR Voice Lines, Local Warning Sirens	- Automated bilingual evacuation route guidance - Location of nearest open, powered cyclone shelter - Transparent relief entitlement confirmation - Missing family member locator queries

2.2 Physical Infrastructure: Hubs, Warehouses & Shelters

495 UPAZILA DISASTER OPERATIONS HUBS (TACTICAL COMMAND RETROFITS)
Every one of Bangladesh's 495 Upazila Parishad complexes will receive a dedicated, turnkey GreenSignal AI Tactical Command Suite. Key physical specifications include: - Dual 65-inch Ultra-HD tactical monitoring screens mirroring live upazila GIS hydrological telemetry, water levels, and alert nodes. 10 kVA Hybrid Solar-Inverter power system with 48V 200Ah Lithium Iron Phosphate (LiFePO4) storage, coupled to a 12 kW sound-attenuated automated diesel generator capable of 14 days grid-independent runtime. - Dedicated multi-band VHF/UHF tactical radio base station linked with the Cyclone Preparedness Programme (CPP) and Bangladesh Police networks. Bangabandhu Satellite-1 (BS-1) VSAT terminal providing continuous broadband Internet connectivity during fiber severance.
1,250 CLIMATE-HARDENED UNION RELIEF BUFFER SILOS & MULTI-PURPOSE SHELTERS
Positioned across vulnerable coastal chars, riverine haor basins, and flash-flood corridors, these facilities safeguard emergency supplies and human lives: - Reinforced 2-Meter Elevated Plinth: Constructed strictly above the 100-year historical flood surge watermark, preventing submersion. - Hermetic Dry Grain Storage Silos: 100-metric-ton airtight silos preserving emergency rice, lentils, and baby food against high humidity, water intrusion, and pest contamination for up to 24 months without spoilage. - Solar Cold-Chain Medical Vaults: WHO PQS-certified medical refrigeration units maintaining 2°C to 8°C for anti-snake venom, cholera vaccines, tetanus toxoid, and essential antibiotics powered by dedicated rooftop PV arrays. - High-Capacity Solar Water Desalination & RO Pods: Autonomous solar-powered reverse osmosis systems producing 5,000 liters/day of certified WHO-standard drinking water directly from saline estuarine or flood sources. - Automated RFID / Barcode Inventory Gates: Real-time scan-in / scan-out tracking of all arriving relief trucks and departing field consignments, transmitting live stock balances to the central PostgreSQL database.

SECTION 3.0: DIGITAL PLATFORM ARCHITECTURE, SOFTWARE CODEBASE & SECURITY

The GreenSignal AI digital platform is built with an enterprise-grade, microservice-oriented full-stack architecture engineered for absolute high-concurrency reliability under catastrophe conditions. All codebase components are strictly typed, horizontally scalable, and sovereignly hosted.

3.1 Full-Stack Software Architecture & Core Codebase Modules

Field Worker PWA Mobile App (TypeScript / Service Workers / IndexedDB):

Operates with 100% offline-first resiliency. Responders can log survivor headcounts, tag localized embankment collapses, and issue emergency medical distress beacons even without cellular reception. Background sync automatically dispatches pending payloads upon establishing any 2G/EDGE or LoRa connection.

Tactical Coordinator GIS Suite (public/index.html & Leaflet Engine):

High-performance interactive geographic dashboard rendering 576+ nationwide upazila district nodes, 503+ relief shelters, and live IoT sensor overlays. Incorporates dynamic mathematical relief allocation algorithms balancing food, drinking water, medical supplies, and temporary shelter kits against real-time displaced populations.

Privacy-Isolated Real-Time Messenger (src/services/messaging.ts):

Engineered on an event-isolated Socket.io architecture. Employs dedicated public broadcast rooms (e.g. upazila-general) for macro situational directives alongside cryptographically segregated 1-on-1 responder channels (e.g. direct-userIdA-userIdB), preventing operational confusion and preserving survivor medical confidentiality.

Relational Database Engine & Spatial Indexing (prisma/schema.prisma):

Rigorous normalized relational schemas governing DistrictNode, User, ReliefCenter, EnvironmentalSensor, DisasterAlert, ShelterAllocation, and ChatMessage with strict foreign-key integrity, ACID transactions, and spatial B-tree indexing for sub-millisecond query latency.

3.2 Core Database Entities & Data Governance

Entity Model	Key Attributes	Functional Role & Data Integrity Constraint
DistrictNode	id, name, division, riskLevel, coordinates	Represents all 64 districts & 495 upazilas; root spatial hierarchy node.
ReliefCenter	id, districtId, capacity, stockFood, stockWater	Maintains live grain, water, and medical inventory balances in real time.
EnvironmentalSensor	id, districtId, type, reading, status, battery	Stores telemetry from radar water gauges, rain gauges, and piezometers.
DisasterAlert	id, districtId, severity, title, active, timestamp	Automated event triggers pushed via WebSocket, Cell Broadcast & SMS.
ShelterAllocation	id, reliefCenterId, allocatedBy, peopleHoused	Tracks survivor occupancy quotas to prevent life-threatening overcrowding.
ChatMessage	id, senderId, receiverId, channelId, message	Stores historical audit log of all tactical responder and command communications.

3.3 National Data Sovereignty & Cyber-Security Framework

- 100% In-Country Data Sovereignty: All database instances, telemetry logs, and citizen records strictly reside inside Bangladesh within the Tier-IV National Data Center. No operational disaster data is routed through foreign cloud jurisdictions.
- Cryptographic Security Standards: Enforces TLS 1.3 for all data-in-transit and AES-256 for all databases, backup snapshots, and local client caches. Responders authenticate via secure JWTs backed by bcrypt password hashing with unique salts.
- Anonymized Citizen Entitlement Tokens: Relief allocation utilizes zero-knowledge HMAC tokens linked to national identification numbers (NID); no plaintext biometric or citizen identification files are stored on field mobile devices.
- Role-Based Access Control (RBAC): Strict segregation of roles (Admin, National Director, District Coordinator, Upazila Lead, Field Worker). Privileged administrative actions are logged to immutable, append-only security ledgers.
- Regular Penetration Testing & CIRT Compliance: Fully audited against the National Cyber Security Act and aligned with BGD e-GOV CIRT vulnerability mitigation protocols.

SECTION 4.0: NETWORK TOPOLOGY & MULTI-AGENCY TELECOMMUNICATIONS

To withstand Category 4/5 cyclone landfalls and catastrophic monsoonal inundations where commercial telecommunications networks routinely collapse, GreenSignal AI integrates a tri-redundant network architecture combining Tier-IV National Data Center hosting, private telco APNs, Bangabandhu Satellite-1 (BS-1), and solar LoRaWAN mesh gateways.

Tier-IV National Data Center (Primary Node)
Located at Bangladesh Computer Council (BCC), Kaliakoir, Gazipur. Features dual 2N redundant electrical feeds, earthquake-isolated seismic dampers, automated FM-200 fire suppression, and guaranteed 99.995% physical availability.

Tier-III Disaster Recovery Center (Hot Standby)
Established at Jashore Software Technology Park. Continuous asynchronous block replication ensures a Recovery Point Objective (RPO) of < 5 seconds and an automated Recovery Time Objective (RTO) of < 15 seconds in the event of primary center isolation.

BDIX Edge & Anycast Peering Fabric
Directly interconnected with the Bangladesh Internet Exchange (BDIX) peering fabric across Dhaka, Chattogram, and Sylhet. Bypasses international submarine cable transit loops, delivering sub-15ms domestic network latency and absorbing DDoS floods up to 500 Gbps.

Kubernetes Cluster & Redis Pub/Sub Grid
Dynamically autoscale worker pods from a baseline of 32 to 512+ compute instances within 180 seconds of disaster alert issuance. A distributed Redis Enterprise Cluster sustains 1,500,000 concurrent persistent WebSocket connections without dropped packets.

4.2 Resilient Multi-Channel Emergency Telecommunications

Channel	Coverage & Providers	Latency & Bandwidth	Disaster Survivability & Role
Dedicated Private APN (4G LTE / 5G)	Grameenphone, Robi, Banglalink, Teletalk	Latency: < 35 ms Speed: 10–50 Mbps	Prioritized QCI-1 data channels for responders. Zero-rated data billing subsidized by MoDMR.
Bangabandhu Satellite-1 (BS-1 Ku/C-Band VSAT)	Bangladesh Satellite Company Ltd (BSCL)	Latency: ~550 ms Speed: 20 Mbps Uplink	Primary backup for 120 isolated coastal char islands (Sandwip, Hatiya, Manpura) during fiber severed.
Solar LoRaWAN Mesh (868 / 915 MHz)	Autonomous Upazila Mast Gateways	Latency: 1–3 sec Speed: 5–50 kbps	Zero-carrier dependency. Broadcasts low-bandwidth sensor telemetry & SOS packets over 15 km line-of-sight.
BTRC Cell Broadcast (SMPP High-Volume)	All 4 Mobile Operators (Direct Tower Broadcast)	Broadcast speed: 100,000 SMS / second	Geofenced emergency push notifications delivered to all mobile handsets without requiring prior app installation.

AUTOMATED NETWORK FAILOVER PROTOCOL (TRI-TIER RECOVERY)
In the event of a terrestrial commercial telecommunications outage, the system initiates an instantaneous, rule-based cascading failover:
1. Detection: Edge monitoring detects loss of 4G APN ping responses for > 15 seconds.
2. Secondary Engagement: Upazila command centers switch traffic dynamically to the Bangabandhu Satellite-1 Ku-band VSAT terminal.
3. Autonomous LoRa Mesh Activation: Field sensors and handheld volunteer terminals switch to the decentralized 868 MHz LoRa mesh, routing emergency distress packets hop-by-hop to the nearest operational satellite uplink node.

SECTION 5.0: AUTOMATED ENVIRONMENTAL SENSOR NETWORK & IOT DATASHEET

Reliable early warning demands high-precision, uncompromised real-world ground truth. GreenSignal AI deploys a national grid of 2,500+ industrial-grade telemetry stations across critical river bends, barrage sluice gates, coastal polders, and flash-flood catchment zones across Bangladesh.

5.1 Comprehensive Environmental Sensor Specifications

Sensor Category	Benchmark Model	Technical Specifications	Interface / Protocol	Unit
Radar River Water Level Sensor	OTT RLS / Vega VEGAPULS C21	Non-contact 26 GHz radar, ±2 mm accuracy, 0–35 m range. Unaffected by flood debris or sediment siltation.	SDI-12 / Modbus RS-485 12V DC, IP68 hermetic	850
Automated Rain Gauge (ARG)	Campbell Scientific TE525MM	Tipping bucket mechanism, 0.1 mm orifice resolution, dual-reed switch, anti-clog debris filter, stainless steel funnel.	Pulse output / Reed switch Gold-plated contacts	650
Embankment Pore Piezometer	RST Instruments VW2100	Vibrating wire transducer, 0–500 kPa range, 0.025% F.S. resolution. Detects subsurface pressure surges preceding levee breach.	Frequency signal Internal thermistor	400
Tidal & Storm Surge Gauge	SonTek Argonaut-XR / Aanderaa	Acoustic doppler velocity + piezo-resistive hydrostatic pressure, 0–20 m depth, integrated marine anti-fouling copper cage.	RS-232 / SDI-12 Submersible titanium	300
Ultrasonic Compact Weather Station	Vaisala WXT536 Multi-Sensor	Ultrasonic wind speed (0–60 m/s) and direction, barometric pressure (600–1100 hPa), ambient air temp (-40°C to +60°C), relative humidity.	Modbus / SDI-12 / NMEA Zero moving parts	300

5.2 Industrial Remote Telemetry Unit (RTU) Engineering

Every environmental sensor connects to a sovereign, ruggedized industrial RTU engineered specifically for tropical delta environments:

- Enclosure: Die-cast marine-grade aluminum housing, IP68 rated, hermetically sealed with Gore-Tex desiccant pressure equalization valves.
- Processing Unit: Ultra-low-power 32-bit ARM Cortex-M4 microcontroller running FreeRTOS with hardware cryptographic engine and watchdog supervisory circuits.
- Dual-SIM Comms Modem: Integrated 4G LTE Cat-M1 / NB-IoT modem with automatic cellular operator handover (Grameenphone/Robi/Teletalk) and secondary 868 MHz LoRa transceiver.
- Autonomous Solar Power Bank: 40W monocrystalline high-efficiency solar panel paired with a 24Ah 12.8V Lithium Iron Phosphate (LiFePO4) battery pack with intelligent Battery Management System (BMS). Guarantees 21 consecutive days of autonomous operation under zero sunlight.
- Mast & Civil Infrastructure: 5-meter hot-dip galvanized steel tower anchored to concrete reinforced foundation pads designed to withstand 260 km/h cyclonic gusts.
- Lightning & Surge Protection: Certified IEC 61643-11 Class II heavy-duty surge arrestors with copper earth spike achieving < 5 ohms ground resistance.

NATIONWIDE SITING & GEO-CORRIDOR ALLOCATION

- Haor & Flash-Flood Basin (800 Stations): Sunamganj, Sylhet, Netrokona, Habiganj, Kishoreganj—monitoring mountain runoff from Meghalaya.
- Major River Embankment Corridors (850 Stations): Kurigram, Gaibandha, Bogura, Sirajganj, Jamalpur along the Jamuna/Brahmaputra and Padma axes.
- Coastal Tidal & Cyclonic Belt (850 Stations): Barguna, Patuakhali, Bhola, Satkhira, Cox's Bazar, Chittagong coastlines and offshore island chars.

SECTION 6.0: COMPREHENSIVE FINANCIAL BUDGET BREAKDOWN (100.00 CRORE BDT)

The financial framework has been rigorously structured to conform strictly with the Public Procurement Act (PPA-2006) and Public Procurement Rules (PPR-2008). The total turnkey project allocation is precisely capped at BDT 100.00 Crore (One Hundred Crore Taka), spanning a 36-month execution phase and 36-month operational warranty.

6.1 Itemized Capital (CapEx) & Operational (OpEx) Allocation Matrix

Expenditure Head & Scope	Key Deliverables & Bill of Quantities	CapEx (Cr)	OpEx (Cr)	Total (Cr BDT)
1. Automated IoT Sensor Network & Hydrometric Telemetry	2,500+ sensor stations, OTT radars, Campbell rain gauges, RST piezometers, solar RTUs, 5m galvanized masts, civil foundations.	25.50	2.50	28.00 Cr
2. Physical Upazila Hubs, Union Warehouses & Cold Storage	495 Upazila Hub retrofits (dual 65" screens, 10 kVA solar inverters, generators), 1,250 Union relief silos, cold-chain vaccine vaults, 5,000 L/d RO plants.	23.00	2.00	25.00 Cr
3. Software Engineering, Tactical GIS Suite & Mobile PWAs	Offline PWA for field workers, Coordinator GIS tactical suite, National Ministry Portal, AI predictive hydrology models, QR/biometric verification.	11.50	2.50	14.00 Cr
4. Data Center Hosting, Cloud, BDIX & Telecom Infrastructure	Tier-IV BCC hosting, Jashore DR center, BDIX peering fabric, 4-operator private APN zero-rated responder bundles, 120 BS-1 satellite terminals.	4.50	7.50	12.00 Cr
5. Nationwide Human Resource Training & Capacity Building	Comprehensive digital literacy training for 76,000+ CPP unit leads, 495 Upazila Committees, simulation drills with Armed Forces Division.	3.00	8.00	11.00 Cr
6. Quality Assurance, Security Audits & Spares Reserve	BGD e-GOV CIRT security audits, ISO 27001 compliance, 15% hardware spares stockpile, unforeseen price escalation contingency reserve.	6.00	4.00	10.00 Cr
TOTAL MANDATED PROJECT OUTLAY (100.00 CRORE CAP)		73.50 Cr	26.50 Cr	100.00 Crore BDT

6.2 Economic Feasibility & Projected Socio-Economic Returns

The implementation of GreenSignal AI generates an overwhelmingly positive benefit-cost ratio (estimated at 11.4 to 1) for the national economy over a 10-year operational horizon:

- **Avoided Agricultural & Relief Spoilage:** Preservation of an estimated BDT 450+ Crore worth of standing crops, livestock, and emergency relief food grains across the 1,250 hermetic Union silos over 5 disaster cycles.
- **Elimination of Relief Leakage:** Transitioning from manual paper lists to digital QR/biometric verification is projected to curb relief misdirection and duplication by > 98%, saving BDT 85 Crore in administrative overheads.
- **Rapid Post-Disaster Economic Recovery:** Autonomous water RO purification and cold-chain vaccine preservation reduce post-flood waterborne disease outbreaks (cholera, dysentery) by an estimated 74% in flood-isolated unions.
- **Life Safety Index:** Advance warning of 3 hours in flash-flood zones and 12 hours in storm-surge belts will virtually eliminate unheralded drownings and structural entrapment casualties.

SECTION 7.0: IMPLEMENTATION ROADMAP, RISK MITIGATION & SIGN-OFF

Execution of this nationwide 100.00 Crore BDT program is structured across three disciplined phases over a 36-month timeline, followed by long-term warranty and joint defense simulation drills.

7.1 36-Month Nationwide Phased Rollout Schedule

Phase 1 (Months 01–06): Inception, Architecture & 25-Upazila Pilot

Establish primary Tier-IV BCC cloud cluster and private 4-telco APN. Deploy 25-Upazila flagship pilot across high-risk pilot zones: Sunamganj (flash floods), Kurigram (riverine erosion), and Barguna (cyclonic storm surges). Calibrate AI predictive hydrology algorithms against historical monsoon datasets.

Phase 2 (Months 07–20): Regional Scale & Physical Hub Retrofitting

Retrofit 250 Upazila Disaster Operations Hubs and construct 600 Union Relief Buffer Silos. Deploy 1,500 automated IoT telemetry stations. Onboard and distribute ruggedized PWAs to 40,000 CPP unit leaders. Integrate real-time radar data feeds with BMD and FFWC.

Phase 3 (Months 21–36): Nationwide Saturation & Inter-Agency Handover

Complete remaining 245 Upazila Hubs and 650 Union centers, achieving 100% national coverage. Deploy remaining 1,000 sensor stations. Conduct joint multi-force disaster simulation drills with the Armed Forces Division and BDRCS. Institutional transition to MoDMR permanent operations.

7.2 Key Program Risks & Institutional Mitigation Strategy

Risk Category	Severity	Mitigation Strategy & Safeguards
Field Sensor Vandalism & Theft	Medium	Tamper-detection vibration switches; GPS geofencing; community stewardship contracts with Union Parishads.
Catastrophic Telecom Blackout	High	Dual Ku-band VSAT on Bangabandhu Satellite-1 paired with solar 868 MHz LoRaWAN peer-to-peer mesh.
Low Digital Literacy in Field	Medium	Icon-driven Bengali PWA UI; zero-text voice note reporting; intensive simulation drills with 76,000+ volunteers.

7.3 Formal Ministerial Endorsement & Approval Block

TECHNICAL CONSORTIUM

Prepared & Certified By:
Lead Systems Architect
GreenSignal AI Delivery Group

Signature & Date

DEPARTMENT OF DISASTER MGMT

Reviewed & Recommended By:
Director General
Department of Disaster Mgmt

Signature & Date

MINISTRY OF DISASTER RELIEF

Approved for Allocation:
Secretary / Hon'ble Minister
Ministry of Disaster Mgmt & Relief

Official Seal & Approval Signature